

Collective cooperation without individual fidelity in LLM agents

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Abstract

Large language models (LLMs) are increasingly used as agents in simulations of social systems, yet it remains unclear when their behavior can be interpreted as a faithful proxy for human decision-making. Here we test LLM agents against a direct empirical benchmark: a large-scale networked Prisoner’s Dilemma experiment with human participants. Using the same interaction protocol, payoff structure, and network topologies, we compare nine open-weight LLMs with the human data. The selected model reproduces several macro-level features of cooperation dynamics, including the early decline and later stabilization of cooperation. This aggregate agreement, however, does not extend uniformly to finer levels of behavior. LLM populations underestimate individual-level heterogeneity and generate conditional cooperation patterns that differ from those observed in humans. Adding a fraction of random agents improves some aspects of micro-level agreement, but does not remove the mismatch in decision rules. These findings reveal a macro–micro dissociation in LLM-based social agents: collective outcomes can appear human-like even when the underlying behavioral distributions and mechanisms are not. They suggest that validating LLM agents as human surrogates requires comparisons across aggregate dynamics, individual heterogeneity, and context-dependent decision rules, rather than outcome-level agreement alone.

Keywords: Large Language Models; Networked Social Dilemmas; Prisoner’s Dilemma; Conditional Cooperation; Behavioral Heterogeneity

1 Introduction

Large language models (LLMs) are increasingly used as proxies for human agents in simulations, evaluation settings, and collective decision-making environments[1–3]. This development is part of a broader shift in which language models are no longer treated only as systems that generate text, but also as agents whose behavior can be elicited, measured, and compared with that of humans. Recent work has extended this approach to experimental and agent-based settings, where LLM agents are used to simulate human-like behavior in social and strategic contexts [1, 4].

The appeal of this approach is clear. If LLM agents could reproduce human behavior at scale, they would provide a powerful tool for computational social science, behavioral experimentation, and the construction of social simulations. Yet this promise also raises a basic validation problem. Agreement with human outputs does not necessarily imply that a model reproduces the behavioral patterns or decision processes that generated those outputs. Several recent studies point in this direction. LLMs may reconstruct users through compressed behavioral representations that introduce systematic biases in social simulations [5]; they may align with human evaluative judgments while relying more heavily on lexical associations and statistical priors than on contextual reasoning [6]; and LLM-generated text can display statistical signatures, such as higher structural regularity and compressibility, that differ from human language production [7]. These observations suggest that apparent behavioral realism may be shallow if it is assessed only at the level of visible outcomes.

This issue is especially important in social simulations and game-theoretic environments, where LLM agents are increasingly used as substitutes for human participants. LLMs can participate effectively in repeated strategic interactions, particularly in self-interested settings such as the iterated Prisoner’s Dilemma family, while showing weaker performance in coordination problems [8]. Other work has shown that LLM agents can develop collective conventions and shared behavioral patterns without explicit programming [9]. At the same time, these agents remain distinguishable from humans in interactive settings: in online debates, for example, LLM agents can remain on topic and blend into conversations, while still being perceived as less convincing and less confident than human interlocutors [4]. Related studies show that LLM outputs can approach human benchmarks in some evaluative tasks and can be systematically shaped through prompting and fine-tuning [10, 11].

A further complication is that current LLMs are not primarily optimized to reproduce human behavior in experiments. They are generally trained and aligned to function as helpful assistants in human-oriented tasks [12]. This distinction matters: a model that is useful, compliant, or socially desirable need not be a faithful model of human decision-making. Alignment procedures may also compress heterogeneous preferences into narrower sets of high-probability behaviors, as shown by Slocum et al. [13], with related empirical evidence of mode collapse in aligned language models [14]. These concerns are consistent with recent commentaries emphasizing the lack of empirical validation against real human behavioral data in LLM-based agent models [15], and with calls to calibrate scientific claims from LLM social simulations to the strength of robustness checks and empirical benchmarks [16].

Here, we evaluate whether LLM agents can function as behavioral surrogates for humans in repeated networked social dilemmas. We use as a benchmark the large-scale Prisoner’s Dilemma experiment of Gracia-Lázaro et al. [17], in which human cooperation converged to similar levels across network topologies, challenging the theoretical expectation that heterogeneous networks should promote cooperation. Using the same network structures and interaction protocol, we compare nine open-weight LLMs operating as autonomous agents in repeated Prisoner’s Dilemma games. Our aim is not to ask whether LLMs play the game better than humans, but whether they reproduce the empirical temporal dynamics, individual heterogeneity, and conditional behavioral patterns observed in the experiment.

This design allows us to test behavioral similarity at several levels. At the macro level, we ask whether LLM agents reproduce the aggregate evolution of cooperation. At the micro level, we ask whether they reproduce the distribution of individual cooperation propensities. At the decision-rule level, we ask whether they reproduce conditional cooperation, namely, the way in which participants adjust their actions to the previous behavior of their neighbors. We find that the selected LLM reproduces several macro-level regularities, including the early decay and later stabilization of cooperation. However, this aggregate agreement does not fully extend to micro-level signatures: simulated populations underestimate behavioral heterogeneity and generate conditional cooperation rules that differ quantitatively from the empirical data. These results reveal a macro–micro dissociation in LLM-based social agents, with direct implications for the validation of machine behavior and for the use of LLM agents as human surrogates in experimental social science.

2 Results

We present the results in three steps. We first describe the empirical benchmark and the simulation framework. We then assess whether LLM agents reproduce aggregate cooperation dynamics, before turning to individual heterogeneity and conditional cooperation. This ordering follows the central validation question of the study: whether agreement at the collective level extends to finer behavioral levels.

2.1 Empirical benchmark and simulation framework

To evaluate whether LLM agents reproduce human cooperation dynamics, we simulated a repeated multi-agent Prisoner’s Dilemma on network structures derived from a human behavioral experiment. The simulation framework consists of an orchestration layer that manages autonomous agents, each controlled by an LLM. This work builds on the experimental design of Gracia-Lázaro et al., who examined human cooperation in networked Prisoner’s Dilemma games. The original study found that human cooperation converged to similar steady-state levels across network topologies, challenging theoretical expectations that heterogeneous networks should promote cooperation. Using the same network structures and temporal dynamics as in the original study, we test whether LLM-based agents reproduce the empirical cooperation patterns or instead follow behavior closer to classical game-theoretic predictions.

The simulation environment is a repeated multi-agent game in which each network node is occupied by an LLM-driven agent. The simulation proceeds in discrete rounds that mirror the original experiment by Gracia-Lázaro et al. In each round, agents receive a prompt summarizing the previous round, including their neighbors’ choices and normalized payoffs. Agents then return their current choice: *cooperate* or *defect*. The underlying incentive structure is a Prisoner’s Dilemma where payoffs are computed pairwise. The total payoff for an agent is the sum of these pairwise interactions across all their network connections. The specific values used to drive the agents’ goal of maximizing “ECUs” (Experimental Currency Units) are detailed in Table 1. As noted by Gracia-Lázaro et al., the payoff values follow Grujic et al (2010) [18]. In this specific configuration, the rewards are designed such that cooperation is theoretically expected to reach a high level [17, 18]. Implementation details, model descriptions, and statistical procedures are provided in the Methods.

Table 1 Pairwise payoff matrix (ECUs per neighbor).

Your Choice / Neighbor Choice	Cooperate	Defect
Cooperate	7	0
Defect	10	0

2.2 Macro-level cooperation dynamics

We first ask whether different LLMs produce comparable aggregate cooperation dynamics when placed in the same experimental environment. For this initial benchmark, we fix the interaction structure to a regular lattice network (625 nodes, average degree 4). Figure 1 shows the resulting cooperation trajectories across all models. The tested LLMs span several model families, parameter scales, and training paradigms. Despite identical incentives and network structure, they generate markedly different cooperation regimes. Notably, *qwen3:32b* produces the highest cooperation levels across the simulation horizon. Ablated variants derived from the same base checkpoints also display systematically altered trajectories relative to their originals, indicating that collective outcomes are sensitive to model choice and alignment. Among the models tested, *llama4:16x17b* yields the closest aggregate match to the empirical data. We therefore use this model as the primary agent in the remaining analyses, while treating the initial benchmark as evidence that LLMs should not be regarded as interchangeable behavioral surrogates.

With this LLM fixed for the remaining analyses, we next evaluate the two network structures considered in [18]: a regular lattice and a heterogeneous network (604 nodes, degrees ranging from $k = 2$ to $k = 16$). Fig. 2A shows the temporal evolution of cooperation for one realization of the dynamics. As in the empirical experiment, both structures yield similar cooperation levels that overlap closely over time. Compared to the empirical data, the simulated trajectories align more closely in later iterations, though they remain noisier overall.

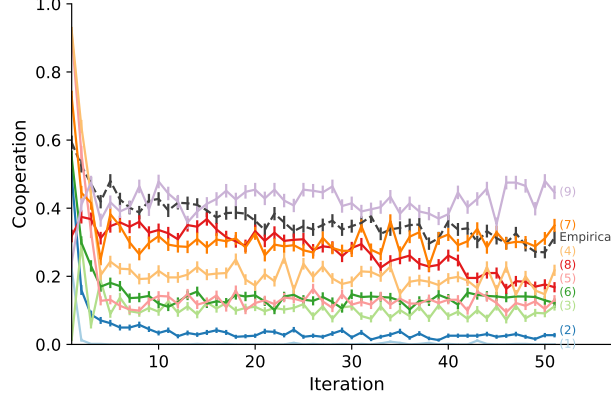


Fig. 1 Comparison of cooperation outcomes across LLM models. The numbers shown on the plot correspond to the following models: (1) BlackHillsInfoSec llama-3.1:8b-abliterated, (2) deepseek-llm:67b, (3) deepseek-llm:7b, (4) krith meta-llama-3.1:70b-instruct-abliterated IQ4 XS, (5) llama3.1:70b, (6) llama3.1:8b, (7) llama4:16x17b, (8) smolLM2.1:7b, and (9) qwen3:32b. Experiments performed for the Lattice network. The solid lines represent the mean cooperation rate per round (empirical) or per iteration (simulation), and the error bars represent the standard error.

To perform a thorough comparison, we also test the control cases, which consist of simulations on a time-varying network structure. Specifically, connections are randomly rewired at each iteration while preserving the original degree sequence. To match the original study, we do not generate a new rewiring sequence; instead, we use the exact sequence of network connections reported in [18], which was originally generated at random. Henceforth, we refer to the fixed-network simulations as the *networked* condition, and to the time-varying simulations as the *control* condition. Figure 2F shows the resulting cooperation trajectories over time. Notably, the heterogeneous control case more closely matches the empirical data, whereas in the lattice control case, agents tend to cooperate less. As in the networked simulations, the LLM-based trajectories remain noisier overall.

Upon closer inspection of the cooperation histograms, we observe that the empirical experiments yield substantially broader distributions than those of the simulations under the networked condition. Specifically, the empirical lattice case (Fig. 2B) has mean cooperation $\mu = 0.368$ with standard deviation $\sigma = 0.170$, while the empirical heterogeneous case (Fig. 2C) has $\mu = 0.405$ and $\sigma = 0.158$. By contrast, the simulated lattice network (Fig. 2D) produces a more concentrated distribution ($\mu = 0.291$, $\sigma = 0.069$), whereas the simulated heterogeneous network (Fig. 2E) yields $\mu = 0.314$ and $\sigma = 0.148$. While the lattice simulation more closely reproduces the empirical mean, the heterogeneous simulation better captures the observed dispersion and therefore provides the closer overall match.

A similar pattern is observed in the *control* condition. The empirical lattice control (Fig. 2G) exhibits $\mu = 0.292$ and $\sigma = 0.209$, and the empirical heterogeneous control (Fig. 2H) shows $\mu = 0.322$ and $\sigma = 0.209$, again indicating wide distributions. In contrast, the simulated lattice control (Fig. 2I) remains narrowly concentrated ($\mu = 0.238$, $\sigma = 0.047$), while the simulated heterogeneous control (Fig. 2J) produces higher

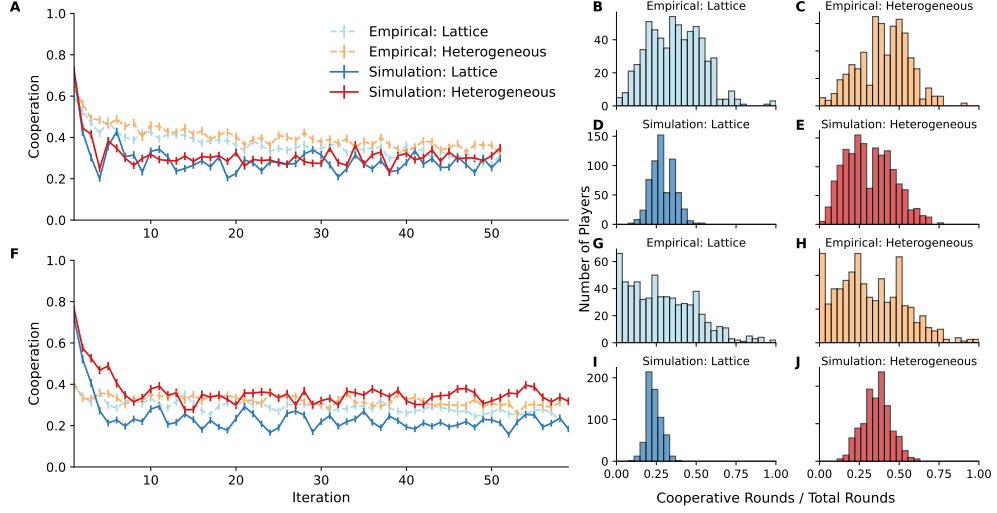


Fig. 2 Evolution of cooperation over iterations. Dashed lines show empirical results from the behavioral experiment, while solid lines show simulation results. The solid lines represent the mean cooperation rate per round (empirical) or per iteration (simulation), and the error bars represent the standard error. Panels A-E show the results for the networked experiment, and Panels F-J show the results for the control case.

cooperation ($\mu = 0.361$) but still substantially reduced variability ($\sigma = 0.089$). Thus, although the heterogeneous control simulation better matches the empirical mean cooperation level, both simulated control cases underestimate the dispersion observed in human behavior.

To evaluate the fidelity of the simulated temporal dynamics, we used metrics to compare simulated and empirical cooperation trajectories. In addition to measuring point-by-point error with Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), we used an autoregressive model (AR(1)) to evaluate the persistence of cooperation. The difference in persistence coefficients (Diff_ϕ) indicates whether simulated and empirical trajectories exhibit similar temporal persistence (i.e., a form of aggregate “memory” in the cooperation series). The difference in implied mean increments (Diff_μ) identifies systematic biases in the average rate of change. Finally, to measure the synchrony of cooperation trends, we calculated the Pearson correlation coefficient for the raw levels (r) and the first differences (r^Δ), ensuring that the simulated fluctuations were synchronized with the empirical observations. To ensure that these results were not due to random chance, we established statistical significance using a Monte Carlo null model that preserves marginal distributions while removing temporal structure.

Table 2 shows that the simulations reproduce the overall shape of the empirical cooperation trajectories, but still deviate systematically in temporal structure. Point-wise errors (RMSE and MAE) are lowest in the *heterogeneous control* condition and highest in the *heterogeneous networked* condition. The AR(1) persistence comparison shows that lattice simulations differ strongly from humans in decision “memory,”

with large and significant Diff_ϕ values (0.217 and 0.464), whereas the heterogeneous networked case shows the closest match in persistence ($\text{Diff}_\phi = 0.071$). Differences in implied long-run means remain small across all conditions ($\text{Diff}_\mu \leq 0.004$), suggesting limited bias in average cooperation levels. Finally, correlations in raw cooperation levels are consistently moderate and significant, but correlations in first differences are weaker, indicating that the simulations capture broad trends better than round-to-round fluctuations.

Experiment	RMSE	MAE	Diff_ϕ	Diff_μ	r	r^Δ
lattice	0.098**	0.083**	0.217***	0.002	0.612***	0.276*
lattice control	0.089***	0.074*	0.464***	0.002***	0.583***	0.214
heterogeneous	0.107**	0.096	0.071	0.002	0.622***	0.151
heterogeneous control	0.079***	0.051	0.359	0.004*	0.440***	0.362***

Table 2 Trajectory errors, autoregressive comparison, and correlation. RMSE and MAE measure point-by-point deviations between empirical and simulated cooperation rates. Diff_ϕ denotes the absolute difference between the AR(1) persistence coefficients, and Diff_μ denotes the absolute difference between the implied long-run means of the simulated and empirical series. r denotes the Pearson correlation coefficient between empirical and simulated trajectories. For RMSE, MAE, Diff_ϕ , and Diff_μ , statistical significance is assessed using the Monte Carlo null model described in the text. For r , significance refers to the null hypothesis of zero correlation ($H_0 : r = 0$). *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$, and no symbol indicates that the result is not statistically significant.

In addition to the results presented, we tested three different “personas” for the LLMs. One about being a student, one about gender and student status, and one about gender only. The results were qualitatively unchanged across all persona conditions. Therefore, we only report results using the student background. Furthermore, we tested the same dynamics across multiple independent runs to assess the robustness of our findings. Throughout the simulation, variability in cooperation remained low for both network structures, and trajectory-level statistics were consistent with those reported in Table 2. Specifically, we obtained comparable error magnitudes across runs, with RMSE values of 0.099 for the lattice and 0.105 for the heterogeneous network, and similar correlations with the empirical trajectories, $r = 0.666$ and $r = 0.708$, respectively. These results suggest that stochastic variation does not drive our main findings (see Supplementary Material, Section S3.2.1, for more details).

As an exploratory test of whether non-strategic behavior contributes to human-like variability, we introduced a fraction of random agents. This choice is motivated by qualitative evidence from the original experiment suggesting that some participants may have played without a stable strategy, or even at random [17]. We assign a fraction ρ of nodes to act as *random agents*; these agents do not follow the LLM policy, but choose between cooperate and defect uniformly at random in every round. The remaining fraction $(1 - \rho)$ of nodes is controlled by the LLM-driven decision rule. Based on a systematic sweep over values of ρ (Supplementary Information, Table S3), we use $\rho = 0.2$ as an illustrative hybrid condition. This value balances improved temporal alignment with the empirical cooperation trajectories, particularly in terms

of fluctuation synchrony ($r^\Delta = 0.469$), with preservation of the overall trajectory shape ($r = 0.690$), while avoiding the stronger distortions observed at higher random-agent fractions. Alternative trajectories for other values of ρ are shown in Supplementary Information, Fig. S19.

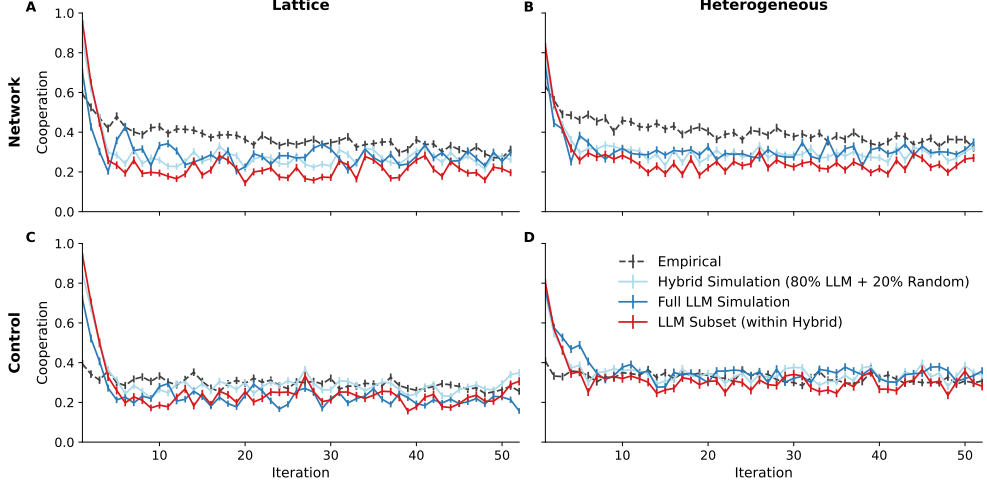


Fig. 3 Evolution of cooperation with a fraction of random agents ($\rho = 0.2$). Panels show cooperation over iterations for (A) lattice, (B) lattice control, (C) heterogeneous, and (D) heterogeneous control network structures. Solid lines denote the mean cooperation rate per round (empirical) or per iteration (simulation), and error bars represent the standard error.

Figure 3 shows that introducing a fraction of random agents ($\rho = 0.2$) preserves the overall qualitative shape of the empirical trajectories, with cooperation rapidly decaying in early rounds and subsequently stabilizing. However, the inclusion of random agents alters the quantitative agreement with human data in a condition-dependent manner. In the lattice network, adding random agents increases the correlation with the empirical trajectory and substantially improves synchrony in short-term fluctuations (r^Δ), although pointwise errors increase slightly. A similar pattern holds for the heterogeneous network, where the hybrid case yields stronger correlation ($r = 0.743$ vs. $r = 0.622$) and higher agreement in first differences ($r^\Delta = 0.343$ vs. 0.151), while maintaining comparable error levels (RMSE = 0.109 vs. 0.107).

In the control scenarios, the effect of random agents is more mixed. In the lattice control case, the hybrid simulation slightly reduces RMSE (0.087 vs. 0.089) and markedly lowers MAE (0.046 vs. 0.074), but decreases correlation with the empirical trajectory ($r = 0.509$ vs. $r = 0.583$). In the heterogeneous control case, the hybrid configuration yields the lowest RMSE across all conditions (0.070), but also produces a weaker correlation ($r = 0.325$), compared to $r = 0.440$ in the scenario without random agents. See Supplementary Material, Table S4, for more details on all the trajectory errors. Together, these results indicate that random agents can improve

alignment with human temporal fluctuations in the networked condition, while in the control condition, they primarily reduce absolute error without consistently improving temporal synchrony.

2.3 Micro-level heterogeneity and decision rules

We next ask whether aggregate agreement is accompanied by individual-level fidelity. To do so, we compare the distribution of cooperation propensities across agents, rather than only the population mean. This distinction is important because two populations can display similar average cooperation while differing substantially in the diversity of individual behaviors. We quantify the distance between empirical and simulated propensity distributions using the first-order Wasserstein distance, $W_1(E, S)$. Lower values indicate closer agreement in behavioral heterogeneity, whereas larger values indicate that the simulated population differs from the empirical one in how cooperation is distributed across individuals. Table 3 summarizes the results across experimental scenarios.

Table 3 shows that the simulations match empirical individual-level heterogeneity with varying accuracy across experimental scenarios. The smallest distance is obtained in the lattice network with mixed populations, where $W_1(E, S) = 0.081$, indicating the closest match to the empirical distribution of cooperation propensities. In contrast, the lattice control condition with a fully LLM-driven population yields the largest discrepancy, $W_1(E, S) = 0.136$. Introducing random agents generally reduces the distance relative to the corresponding fully LLM-driven case, with the strongest improvement observed in the lattice control condition (from 0.136 to 0.095). In the heterogeneous network, however, the mixed-population case slightly worsens the match under the networked condition (from 0.091 to 0.095).

Experiment	$W_1(E, S)$
heterogeneous (100% LLM + 0% Rand.)	0.091
heterogeneous (80% LLM + 20% Rand.)	0.095
heterogeneous control (100% LLM + 0% Rand.)	0.109
heterogeneous control (80% LLM + 20% Rand.)	0.093
lattice (100% LLM + 0% Rand.)	0.101
lattice (80% LLM + 20% Rand.)	0.081
lattice control (100% LLM + 0% Rand.)	0.136
lattice control (80% LLM + 20% Rand.)	0.095

Table 3 Wasserstein distance between empirical and simulated distributions of individual cooperation propensities across experimental scenarios. Lower values indicate better reproduction of behavioral heterogeneity.

For each network condition (lattice, lattice control, heterogeneous, heterogeneous control), we quantified behavioral heterogeneity in the empirical data as the distribution of agent-level cooperation propensities (fraction of cooperative actions per agent). We then generated a parametric null model in which each agent follows an independent

Bernoulli process with success probability equal to the network-level empirical mean cooperation rate, while preserving each agent’s number of rounds. Across ($B=10,000$) null realizations, we computed $W_1(E, \text{Null})$ between empirical and null propensity distributions. We report the mean null distance and a 95% confidence interval of this Monte Carlo distribution, providing a baseline for whether observed heterogeneity departs from what is expected under homogeneous random behavior.

To contextualize these values, we compare the results of Table 3 with the null-model baseline. In the lattice network, the null model yields $W_1(E, \text{Null}) = 0.085$ (95% CI: [0.082, 0.089]), which is close to the best-performing simulation ($W_1(E, S) = 0.081$), indicating that the empirical heterogeneity in this condition can largely be explained by variation consistent with independent behavior around the population mean. A similar pattern is observed in the lattice control condition, where $W_1(E, \text{Null}) = 0.125$ (95% CI: [0.123, 0.128]) is comparable to the full-LLM case ($W_1(E, S) = 0.136$), while the mixed simulation ($W_1(E, S) = 0.095$) falls below the null baseline. In the heterogeneous network, the null distance is smaller ($W_1(E, \text{Null}) = 0.074$, 95% CI: [0.071, 0.077]) than both simulation cases ($W_1(E, S) = 0.091$ and 0.095), suggesting that the simulated populations exhibit less empirical-like heterogeneity than expected under the null benchmark. Finally, in the heterogeneous control condition, the null distance is again high ($W_1(E, \text{Null}) = 0.126$, 95% CI: [0.123, 0.129]), whereas both simulation scenarios yield lower distances ($W_1(E, S) = 0.109$ and 0.093), indicating improved agreement with empirical heterogeneity compared to the null expectation.

We then examine a stricter behavioral signature: *conditional cooperation*. This measures the probability that an agent cooperates as a function of the fraction of cooperating neighbors in the previous round, separately according to whether the agent cooperated or defected in the previous round. Conditional cooperation is important because it probes the local decision rule, rather than only the resulting level of cooperation. As in the empirical study, we first analyze the networked condition, where neighborhoods remain fixed. In the human experiment, this relationship is well approximated by a linear trend with distinct responses following cooperation and defection. Across most simulated scenarios, we observe a qualitative dependence on neighborhood cooperation, but the responses are quantitatively different and often steeper than in the human data. The “after defection” response deviates most strongly from the empirical pattern in the lattice network, exhibiting an inverted slope in both the full-LLM and hybrid simulations. Moreover, unlike the empirical data, the simulated conditional-cooperation points do not consistently follow an approximately linear relationship, making the linear-fit comparison used in the original study less informative here. We therefore report the full curves in the Supplementary Material, Section S3.1, highlighting an additional discrepancy between LLM-based agents and empirical behavioral patterns.

To extend this analysis, we examine conditional cooperation across network structures and experimental conditions. Figure 4 shows conditional cooperation across experimental scenarios for a single execution of the dynamics. In each case, the curves represent the estimated conditional probabilities of cooperation as a function of neighborhood cooperation in the previous round. We repeated this conditional-cooperation analysis over 20 independent executions of the dynamics and obtained consistent

curves with narrow confidence intervals, indicating that a single execution is sufficient to capture the qualitative conditional-cooperation trends of the simulations (for details see Supplementary Material, Figs. S28 and S29).

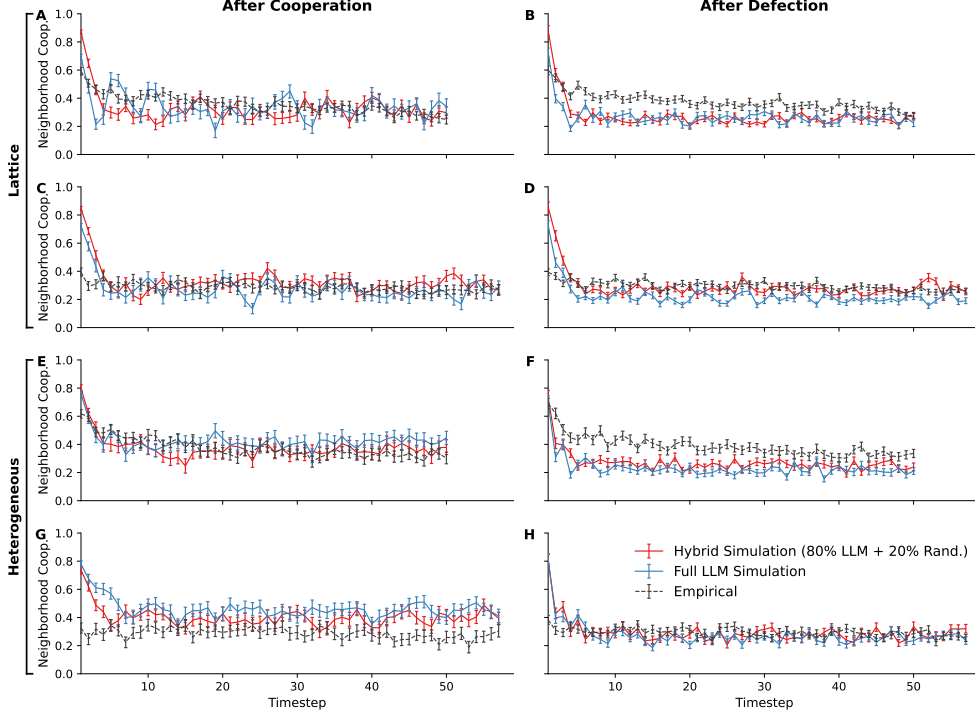


Fig. 4 Conditional cooperation across network structures and experimental conditions. Each panel shows the estimated probability of cooperation as a function of the fraction of cooperating neighbors in the previous round, computed from a single execution of the dynamics. Panels A and B correspond to the lattice network under the control condition, C and D to the lattice network under the networked condition, E and F to the heterogeneous network under the control condition, and G and H to the heterogeneous network under the networked condition. In each case, empirical conditional cooperation patterns are compared with simulations under different agent compositions, including fully LLM-driven and mixed populations with random agents. Error bars indicate 95% confidence intervals.

The quantitative deviations reported in Table 4 are consistent with the qualitative patterns shown in Fig. 4. Across most scenarios, errors are larger in the $z = 1$ case (after cooperation) than in the $z = 0$ case (after defection), reflecting stronger discrepancies in how simulated agents condition their behavior on neighborhood cooperation following a cooperative move. The largest deviations are observed in the lattice network under the networked condition, where the full-LLM population yields $\text{RMSE}_{z=1} = 0.397$ and $\text{MAE}_{z=1} = 0.388$, matching the visibly distorted and non-linear response in the corresponding panel. Introducing random agents reduces these

discrepancies in the lattice networked case ($\text{RMSE}_{z=1}$ decreases to 0.255), although a substantial mismatch remains.

In contrast, the lattice control condition shows comparatively small error after defection ($\text{RMSE}_{z=0} = 0.046$ in the full-LLM case), consistent with the flatter conditional patterns visible under rewiring, while errors after cooperation remain large ($\text{RMSE}_{z=1} = 0.341$). For the heterogeneous network, the mixed population reduces deviations relative to the full-LLM case, particularly in the $z = 1$ regime ($\text{RMSE}_{z=1}$ decreases from 0.355 to 0.232), which aligns with the closer agreement between the empirical and simulated curves observed in the corresponding panels. Overall, these results confirm that while simulations capture the qualitative dependence of cooperation on local neighborhood behavior, the inferred conditional rules remain quantitatively distinct from the empirical patterns, particularly following cooperative actions.

Experiment	$\text{RMSE}_{z=0}$	$\text{MAE}_{z=0}$	$\text{RMSE}_{z=1}$	$\text{MAE}_{z=1}$
lattice (100% LLM + 0% Rand.)	0.237**	0.217***	0.397***	0.388*
lattice (80% LLM + 20% Rand.)	0.180	0.165***	0.255***	0.246*
lattice control (100% LLM + 0% Rand.)	0.046***	0.038***	0.341***	0.337***
lattice control (80% LLM + 20% Rand.)	0.090*	0.086*	0.221***	0.211***
heterogeneous (100% LLM + 0% Rand.)	0.177***	0.154	0.355***	0.320***
heterogeneous (80% LLM + 20% Rand.)	0.161***	0.140	0.232**	0.214
heterogeneous control (100% LLM + 0% Rand.)	0.107**	0.091***	0.217	0.196
heterogeneous control (80% LLM + 20% Rand.)	0.190	0.124	0.173	0.161**

Table 4 Deviation between empirical and simulated conditional cooperation rules across experimental scenarios. Errors are reported separately for cases in which the previous action was defection ($z = 0$) or cooperation ($z = 1$).

*** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$, and no symbol indicates that the result is not statistically significant.

3 Discussion

This study tested whether autonomous LLM agents can serve as behavioral surrogates for humans in a repeated networked social dilemma. The answer is mixed in a revealing way. The selected model, *llama4:16x17b*, reproduces several macro-level features of the human experiment, including the early decline and later stabilization of cooperation observed by Gracia-Lázaro et al. [17]. Yet this aggregate agreement does not imply individual-level fidelity. The simulated populations underestimate important aspects of human heterogeneity and generate conditional cooperation patterns that differ from the empirical decision rules.

This pattern is the central result of the paper. LLM-based agents can reproduce collective cooperation dynamics without fully reproducing the distribution of individual behaviors or the context-dependent mechanisms through which humans adapt their decisions. In other words, the model captures part of the phenomenology of the

experiment, but not all of its behavioral structure. This macro–micro dissociation matters because many proposed uses of LLM agents in social simulation rely implicitly on the assumption that matching aggregate outcomes is sufficient evidence of behavioral realism.

Our model benchmark also shows that LLMs cannot be treated as interchangeable agents. Different model families produce markedly different cooperation regimes, and alignment-perturbed variants derived from related checkpoints can generate systematically different collective dynamics. Model choice is therefore a substantive modeling assumption, not a technical detail. This sensitivity reinforces the need for empirical benchmarking before drawing conclusions from LLM-based social simulations [15].

The micro-level results are especially informative. Human participants display broad distributions of cooperation propensities, whereas the simulated populations often produce narrower or differently shaped distributions. Conditional cooperation provides an even stronger test. Human behavior in the original experiment is characterized by structured responses to the local social environment, with distinct patterns after cooperation and after defection. The LLM agents reproduce some qualitative dependence on neighborhood cooperation, but the inferred rules differ quantitatively, and in some cases qualitatively, from the empirical patterns. Thus, the model does not simply implement a noisy version of the human rule; it appears to reach partially similar aggregate outcomes through different local response functions.

The hybrid simulations with random agents help clarify this point. Introducing a fraction of random decision-makers improves agreement with the empirical data in some dimensions, particularly fluctuation synchrony in the networked condition and distributional agreement in selected cases. This should not be interpreted as a tuned correction that makes the model human-like. Rather, it suggests that stochasticity, inconsistent behavior, or non-strategic play may be necessary ingredients in models of human cooperation. Real participants are not homogeneous optimizers, and a faithful behavioral surrogate may need to represent this diversity explicitly. At the same time, the hybrid model does not remove the mismatch in conditional cooperation, indicating that noise alone is not sufficient to recover human decision mechanisms.

These results have broader implications for the validation of machine behavior. A model may be useful for reproducing an aggregate trajectory while still being inadequate for causal explanation or intervention design. This distinction is particularly important for digital twins, policy simulations, and agent-based models that use LLMs as substitutes for human actors. If the aim is only to forecast a limited aggregate outcome under conditions close to the benchmark, macro-level agreement may be informative. If the aim is to understand how individuals respond to incentives, social context, or interventions, then micro-level heterogeneity and decision-level rules become essential.

Several limitations should be kept in mind. The empirical benchmark comes from a specific experiment, participant pool, and institutional context, and the same comparison should be repeated across other social dilemmas and populations. The simulations also rely on a particular prompt structure and on open-weight models available at the time of the study. Although persona variants and repeated executions did not qualitatively change the results, broader prompt and model robustness remain an

important direction for future work. Finally, current LLMs are typically optimized to follow human instructions and preferences, not to reproduce the full diversity of human behavior in experiments [12]. This may contribute to the reduced behavioral diversity observed here, consistent with forms of model homogenization and mode collapse reported in recent work [13, 14]. More generally, our findings support the view that claims based on LLM social simulations should be calibrated to the strength of empirical validation and robustness checks [16].

Overall, *llama4:16x17b* provides a promising but incomplete surrogate for human behavior in networked social dilemmas. It reproduces robust macro-level cooperation dynamics, but not the full heterogeneity and conditional structure of human strategic decision-making. The main lesson is therefore not simply that LLM agents succeed or fail as human surrogates, but that surrogacy is level-dependent. Validating machine behavior requires asking not only whether artificial agents produce similar outcomes, but also whether they do so through similar behavioral distributions and decision mechanisms.

4 Methods

4.1 Simulation framework

All LLM-based agents were implemented using the CrewAI orchestration framework,¹ which handles prompt construction, model querying, response parsing, and payoff updates at each iteration. Model inference was performed locally using Ollama² as the serving backend, enabling self-contained execution of all agents without external API dependencies. Each agent was mapped to a node in the network and interacted only with the information in its local prompt, which contained its neighbors’ previous-round actions and the corresponding normalized payoffs. Models were queried independently for each agent at each round, ensuring decentralized decision-making consistent with the experimental protocol.

We simulate a repeated Prisoner’s Dilemma game on empirical social networks (using the same network topologies and temporal dynamics as in Gracia-Lázaro *et al.* [17]). Each node (agent) chooses between GREEN and BROWN in each round, corresponding to cooperation and defection, respectively. We use color labels rather than the canonical game actions in the agent-facing prompts to reduce the likelihood that the LLM recognizes the task from familiar Prisoner’s Dilemma terminology and responds on that basis rather than from the local interaction history and payoffs. The game proceeds in discrete rounds. In each round, each agent observes its neighbors’ previous choices and (degree-)normalized payoffs, then chooses an action for the current round. Payoffs were computed pairwise as follows: if both agents choose GREEN, each receives 7 ECUs; if one chooses BROWN and the neighbor chooses GREEN, the BROWN agent receives 10 ECUs, and the GREEN agent receives 0 ECUs; if both choose BROWN, both receive 0 ECUs. An agent’s total payoff in a round is the sum of these pairwise payoffs with all neighbors. A fraction ρ of nodes were designated

¹<https://www.crewai.com/>, accessed March 2026

²<https://ollama.com/>, accessed March 2026

random (i.e., choosing actions uniformly), while all other nodes used the LLM-driven policy.

We also tested conditions with optional backstory cues in the agent prompts (e.g., labeling an agent as a student or specifying a gender, or both); details of these variants are omitted here. Full prompt templates, backstory variants that define the persona, and implementation details, including parsing routines, are provided in the Supplementary Information, Section S1.

4.2 LLM models and framework

We evaluate nine open-weight large language models spanning multiple model families, parameter scales, and training paradigms, including variants of LLaMA [19], DeepSeek [20], Qwen [21], and SmolLM [22]. All models were selected to represent a broad spectrum of open-source LLMs commonly used in agent-based simulations. The evaluated models belong to four open-weight families. LLaMA models are decoder-only transformer language models developed by Meta, available in multiple generations and sizes. In this study, we use both standard instruction-tuned variants (i.e., llama3.1:8b, llama3.1:70b, and llama4:16x17b) and alignment-modified derivatives (i.e., BlackHillsInfoSec llama-3.1:8b-abliterated and meta-llama-3.1:70b-instruct-abliterated IQ4 XS), allowing us to assess whether differences in model scaling and instruction tuning affect emergent cooperative behavior in multi-agent simulations. In particular, we compare standard instruction-tuned LLaMA models with modified “abliterated” variants of the same base architecture, which serve as alignment-perturbed counterparts to evaluate the sensitivity of the results to behavioral steering. SmolLM models are compact open-weight language models designed for efficient inference and deployment. Here we test smolLM2.1:7b. DeepSeek models are open-weight language models released in multiple parameter scales and variants, and we tested two different sizes deepseek-llm:7b and deepseek-llm:67b. Qwen models are multilingual instruction-tuned large language models, with recent variants (e.g., Qwen3) showing strong performance on reasoning and general language understanding benchmarks, here we use qwen3:32b.

4.3 Quantitative analysis

We evaluate whether the simulations reproduce the empirical experiments at two complementary levels: (i) *macro* dynamics (i.e., the time evolution of the fraction of cooperators in the network), and (ii) *micro* behavior (i.e., heterogeneity across individuals and context-dependent decision rules). Full definitions, robustness checks, and statistical testing procedures are reported in the Supplementary Information (Section S2).

4.3.1 Macro-level agreement

For each condition and network, we summarize behavior at iteration t by the cooperation rate

$$X^{\mathcal{D}}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} [a_i^{\mathcal{D}}(t) = C] , \quad (1)$$

where $\mathcal{D} \in \{E, S\}$ denotes empirical (E) or simulated (S) data, $X^{\mathcal{D}}(t)$ is the fraction of cooperative agents, and C means cooperation. We then compare empirical and simulated series using two complementary notions of similarity. First, we compute pointwise trajectory errors (RMSE and MAE). These measures quantify how close simulations are in *level* at each iteration, in which smaller values indicate closer agreement, with RMSE penalizing large deviations more. Second, we compute Pearson correlation between the two trajectories, both on the raw series (r) and on first differences (r^{Δ}). Correlation captures whether simulations reproduce the *temporal pattern*, where co-movement of increases and decreases, even when small level shifts persist.

4.3.2 Macro-level temporal dependence

Trajectory similarity in level or correlation does not guarantee that the simulated process evolves over time in the same way as the empirical one. To compare temporal dependence, we fit a first-order autoregressive model (AR(1)) [23] to the first-differenced series $\Delta X^{\mathcal{D}}(t)$ (stationarity check [24] and the differenced specification are detailed in the Supplementary Information, Section S2). For a series $\Delta X^{\mathcal{D}}_{c,g,r}(t)$, the AR(1) model is given by

$$\Delta X^{\mathcal{D}}_{c,g,r}(t) = \kappa + \phi \Delta X^{\mathcal{D}}_{c,g,r}(t-1) + \varepsilon(t), \quad (2)$$

where κ is a constant, ϕ captures persistence, and $\varepsilon(t)$ is a white-noise error term. We estimate the model separately for the empirical series $X^E_{c,g,r}(t)$ and the simulated series $X^S_{c,g,r}(t)$. The coefficient ϕ summarizes how strongly changes in cooperation propagate from one iteration to the next: values closer to zero indicate rapid adjustment, whereas larger magnitudes indicate more persistent (or anti-persistent) dynamics. We quantify dynamic similarity using the absolute difference $|\phi_S - \phi_E|$, and we additionally compare the implied mean increment of the differenced process, which captures systematic drift in cooperation over time.

4.3.3 Significance via null models

Because the absolute magnitudes of these metrics depend on the scale and length of the series, we assess whether the observed agreement exceeds what would be expected by chance. We construct Monte Carlo null time series that preserve the empirical marginal distribution of cooperation rates while destroying temporal dependence, and we compute two-tailed p -values by comparing the observed metric to its null distribution (see Supplementary Information, Section S2.2.5). For correlations, we test against the standard null hypothesis of zero correlation.

4.3.4 Micro-level mechanisms

We quantify individual behavioral heterogeneity by computing each agent's cooperation propensity over the full time horizon T . For dataset $\mathcal{D}$, this is defined as

$$P_i^{\mathcal{D}} = \frac{1}{T} \sum_{t=1}^T y_i^{\mathcal{D}}(t), \quad (3)$$

where $y_i^{\mathcal{D}}(t) = \mathbf{1}[a_i^{\mathcal{D}}(t) = C] \in \{0, 1\}$ indicates whether agent i cooperated at round t . To test whether simulations reproduce individual-level diversity, we compare the empirical and simulated distributions of $\{P_i\}$ using the first-order Wasserstein distance (W_1). Lower values indicate closer agreement in behavioral heterogeneity, rather than merely similar mean cooperation levels (see Supplementary Information, Section S2.3). To probe context-dependent strategies, we analyze conditional cooperation as a function of (i) an agent’s own previous action and (ii) the fraction of cooperating neighbors in the previous round (for details see Supplementary Information, Section S2.3.4).

Code availability

The codes necessary to reproduce this study can be found here: [To be added when published].

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